Trigonometry in 3D

1 All edges of the following cube are 7 cm long:

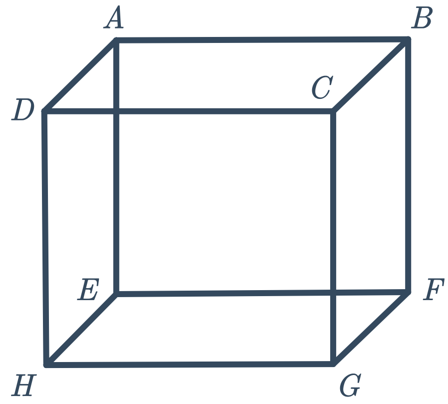
a Find the exact length of the following:

i EG **ii** AG

b Find the following angles to the nearest degree:

i $\angle EGH$ **ii** $\angle EGA$

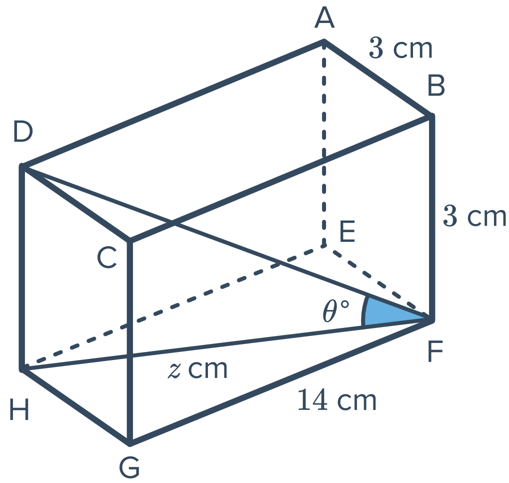
iii $\angle AHG$ **iv** $\angle AGH$



2 A square prism has sides of length 3 cm, 3 cm and 14 cm as shown in the diagram:

a If the diagonal HF has a length of $z\text{ cm}$, find the value of z to two decimal places.

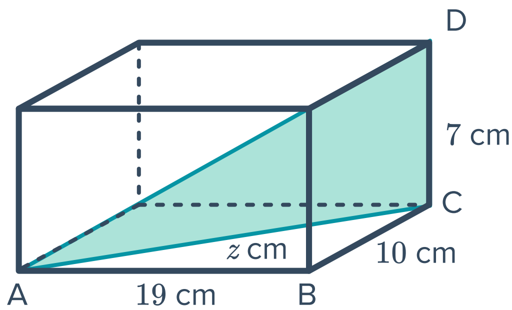
b If the size of $\angle DFH$ is θ°, find θ to two decimal places.



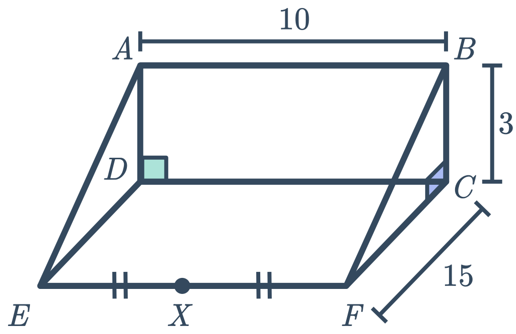
3 The box have a triangular divider placed inside it, as shown in the diagram:

a If $z = AC$, find the value of z to two decimal places.

b Find the area of the divider correct to two decimal places.



4 A triangular prism has dimensions as show[redacted]he diagram:



a Find the size of $\angle AED$, correct to two decimal places.

b Find the exact length of CE.

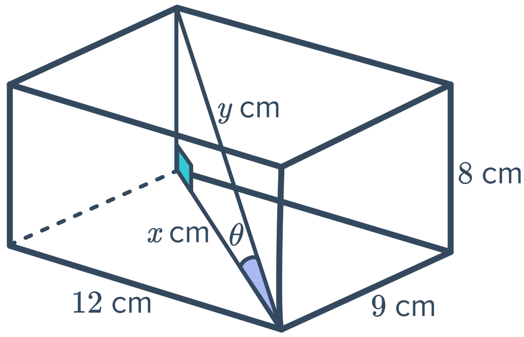
c Find the exact length of BE.

d Find the size of $\angle BEC$, correct to two decimal places.

e Find the exact length of CX.

f Find the size of $\angle BXC$, correct to two decimal places.

5 Consider the given rectangular prism:

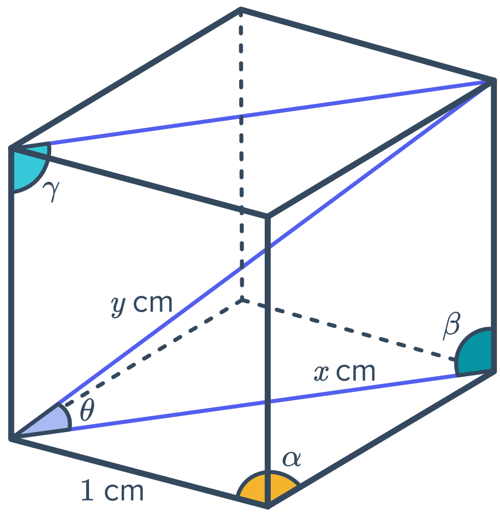


a Find the length x.

b Hence, find the length of the prism's diagonal y.

c Find the angle θ to the nearest degree.

6 Consider the cube as shown:



a Find the size of:

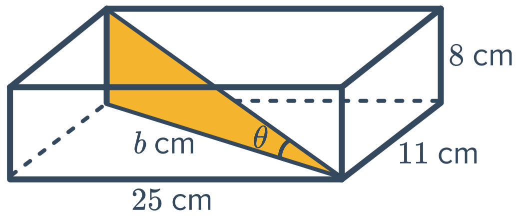
i α **ii** β **iii** γ

b Calculate the exact length of x.

c Calculate the exact length of y.

d Hence, find the angle θ to the nearest degree.

7 A $25\text{ cm} \times 11\text{ cm} \times 8\text{ cm}$ cardboard box c[redacted]s an insert (the shaded area) made of foam:

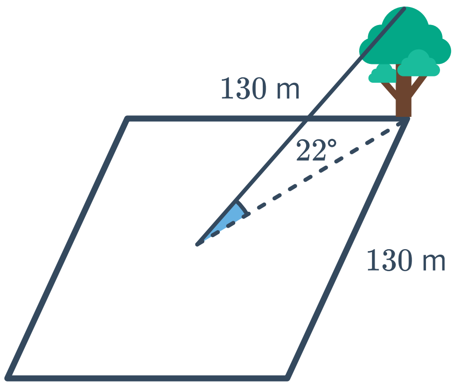


a Find the exact length b of the base of the foam insert.

b Find the area of foam in the insert, rounded to the nearest square centimetre.

c Find the value of θ to the nearest degree.

8 A tree stands at the corner of a square playing field. Each side of the square is 130 m long. At the centre of the field the tree subtends an angle of 22° as shown:



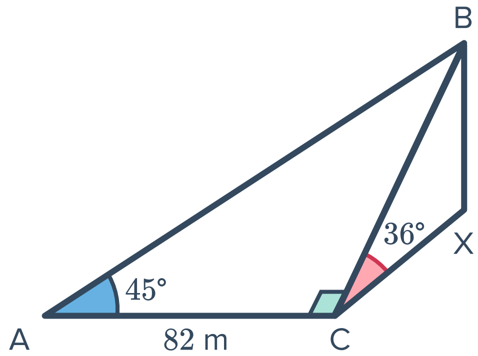
a Find the distance from the tree to the centre of the field. Round your answer to two decimal places.

b Find the height of the tree. Round your answer to two decimal places.

c Find the size of the angle subtended by the tree at the corner of the field opposite the tree. Round your answer to the nearest minute.

d Find the size of the angle subtended by the tree at an adjacent corner of the field. Round your answer to the nearest minute.

9 A, C, and X are three points in a horizontal plane and B is a point vertically above X as shown in the diagram:



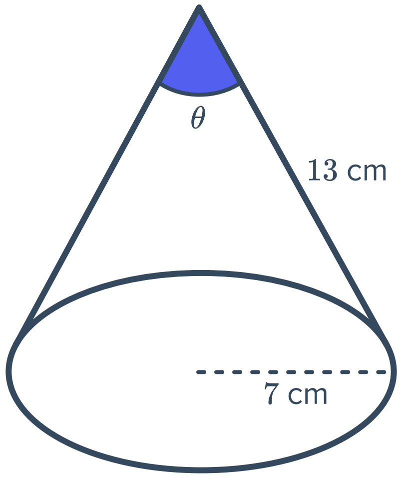
Find the following lengths to the nearest metre:

a BC

b XB

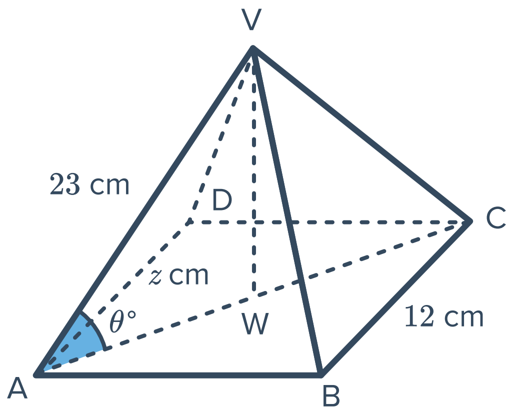
10 A cone has radius 7 cm and a slant height 13 cm:

Find the vertical angle, θ, at the top of the cone in degrees and minutes.



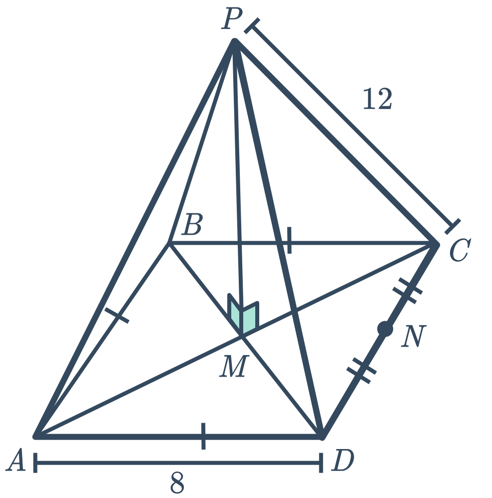
11 The following is a right pyramid on a square base with side length 12 cm. The edge length VA is 23 cm.

- **a** If $z = AW$, calculate the length z.
- **b** If $\theta = \angle VAW$, find the value of θ.



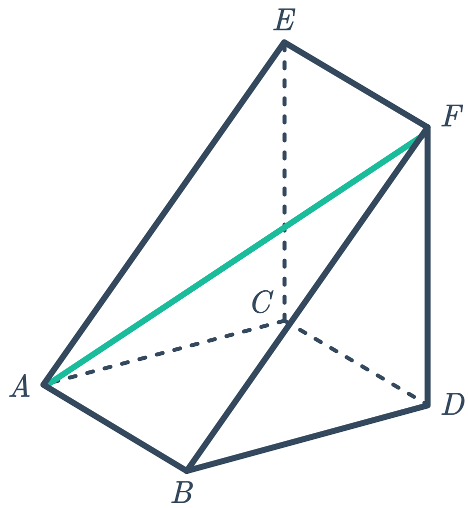
12 The following pyramid has a square base with side length 8 cm, and all slanted edges are 12 cm in length:

- **a** Find the exact length of MD.
- **b** Find the size of $\angle PDM$, correct to two decimal places.
- **c** Find the exact height of the pyramid, MP.
- **d** Find the length of MN.
- **e** Find the exact length of PN.
- **f** Find the size of $\angle PNM$, correct to two decimal places.
- **g** Find the size of $\angle PDC$, correct to two decimal places.



13 This triangular prism shaped box needs a diagonal support inserted as between A and F as shown:

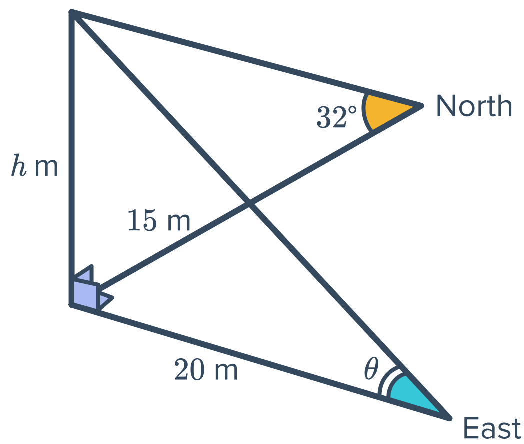
$AB = 19$, $BD = 30$ and $DF = 43$. find the length of AF to two decimal places.



14 From a point $15\ \mathrm{m}$ due north of a tower, the angle of elevation of the tower is 32°.

a Find the height of the tower h. Round your answer to two decimals places.

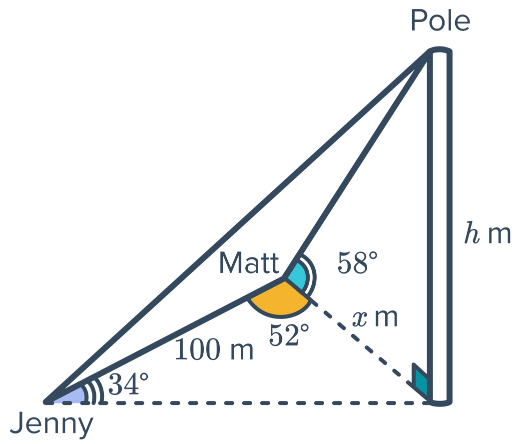
b Find the size θ of the angle of elevation of the tower at a point $20\ \mathrm{m}$ due east of the tower. Round your answer to the nearest degree.



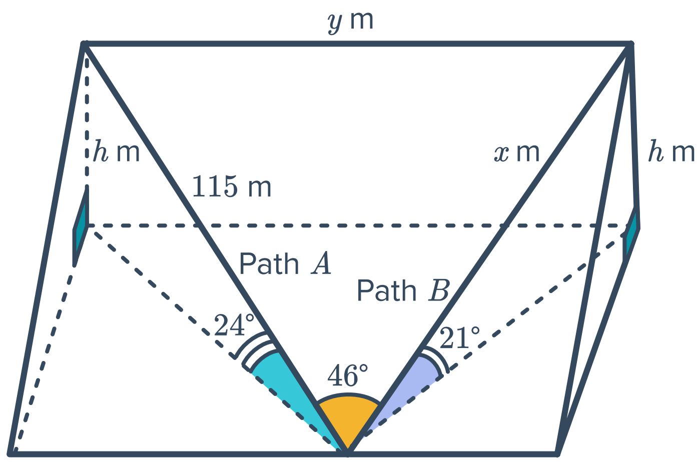
15 A pole is seen by two people, Jenny and Matt:

a Matt is $x\ \mathrm{m}$ from the foot of the pole. Find x to the nearest metre.

b Find the height of the pole h to the nearest metre.



16 Two straight paths to the top of a cliff are i[redacted]ed at angles of 24° and 21° to the horizontal:



a If path A is 115 m long, find the height h of the cliff, rounded to the nearest metre.

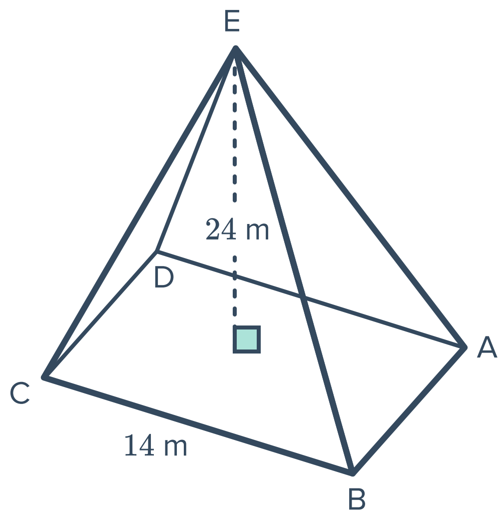
b Find the length x of path B, correct to the nearest metre.

c Let the paths meet at 46° at the base of the cliff. Find their distance apart, y, at the top of the cliff, to the nearest metre.

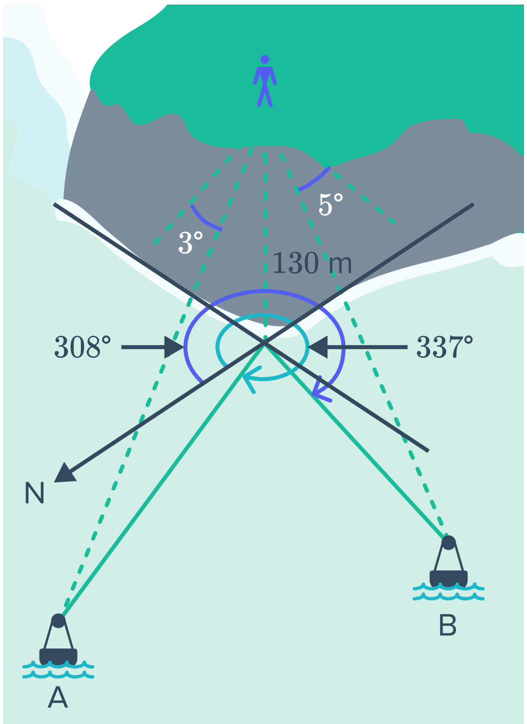
17 A pyramid has a square base with side length 14 m and a vertical height of 24 m:

a Find the length of the edge AE. Round your answer to one decimal place.

b Find the size of $\angle BEA$, to the nearest minute.

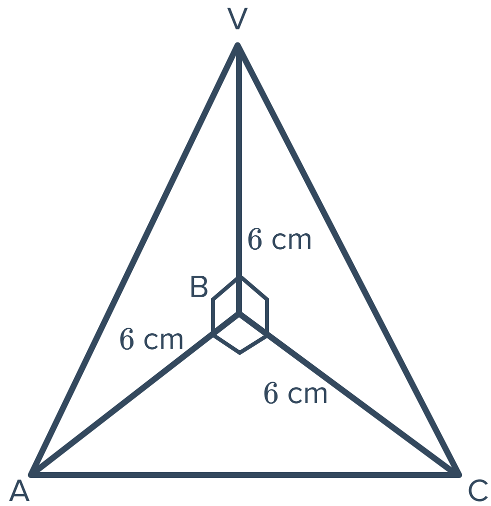


18 Two buoys, A and B, are observed from a lookout at the top of a 130 m high cliff. The bearing of buoy A is 337° and its angle of depression is 3°. The bearing of buoy B is 308° and its angle of depression is 5°:



- **a** Find the distance, to the nearest metre, from buoy A to the base of the cliff.
- **b** Find the distance from buoy B to the base of the cliff, to the nearest metre.
- **c** Find the distance between the two buoys, to the nearest metre.

19 In the tetrahedron shown, the angles $\angle VBC$, $\angle VBA$, and $\angle ABC$ are all right-angles.



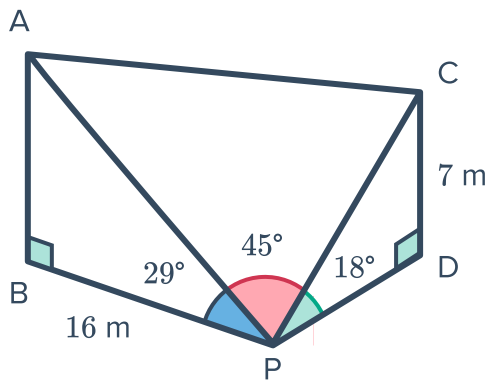
- **a** Find the following distances, to two decimal places:

  i VA **ii** VC **iii** AC

- **b** Find the size of angle $\angle VCA$. Round your answer to the nearest minute.

20 Roald is standing at point P and observes two poles, AB and CD, of different heights. P, B, and D are on horizontal ground:

From P, the angles of elevation to the top of the poles at A and C are 29° and 18° respectively. Roald is 16 m from the base of pole AB. The height of pole CD is 7 m.



- **a** Calculate the distance from Roald to the top of pole CD, to two decimal places.
- **b** Calculate the distance from Roald to the top of pole AB, to two decimal places.

c Calculate the distance between the to
of the poles, to two decimal places.

21 Point A is due north of a tower, and has an angle of elevation to the top of the tower of 51°. Point B is 100 m from point A on a bearing of 120°. The angle of elevation from point B to the top of the tower is 25°.

Find the height of the tower to the nearest metre.

22 A room measures 5 m in length and 4 m in width. The angle of elevation from the bottom left corner to the top right corner of the room is 57°.

a Find the exact distance from one corner of the floor to the opposite corner of the floor.

b Find the height of the room. Round your answer to two decimal places.

c Find the angle of elevation from the bottom corner of the 5 m long wall to the opposite top corner of the wall. Round your answer to two decimal places.

d Find the angle of depression from the top corner of the 4 m long wall to the opposite bottom corner of the wall. Round your answer to two decimal places.

23 A and B are two positions on level ground. From an advertising balloon at a vertical height of 740 m, A is observed in an easterly direction and B at a bearing of 159°. The angles of depression of A and B, as viewed from the balloon, are 30° and 20°, respectively.

a Find the distance from point A to the point directly below the advertising balloon, to the nearest metre.

b Find the distance from point B to the point directly below the advertising balloon, to the nearest metre.

c Find the distance between points A and B, to the nearest metre.